

REVIEW

to the Senior Thesis “Earthquake Magnitude Estimation from Early P-Wave Observations Using a Machine Learning Approach” by Aidarbek Papakov submitted to the Applied Mathematics and Informatics Program of American University of Central Asia in Partial Fulfillment of the Requirements for the Degree of Bachelor of Arts

This thesis addresses one of the key challenges in earthquake early warning systems — accurately estimating earthquake magnitude from only the first seconds of seismic observations. The author proposes a deep learning framework that combines convolutional and recurrent neural networks with classical signal processing techniques, resulting in a robust CNN–LSTM–MLP architecture for early earthquake magnitude prediction from P-wave spectrogram data.

The thesis consists of eight chapters, including an introduction and a conclusion, and a reference list.

Chapter 1 is introduction. Chapter 2 presents the theoretical background of the study by introducing fundamental seismological concepts and reviewing existing earthquake early warning systems based on both traditional and machine learning methods. In Chapter 3, Aidarbek introduces the mathematical foundations of the proposed methodology, including the formal description of signal processing procedures, neural network architectures, optimization techniques, and evaluation metrics. Chapter 4 describes the dataset used in the research and explains the preprocessing and filtration procedures applied during data preparation. Chapter 5 details the proposed CNN–LSTM–MLP architecture, focusing on the convolutional feature extractor, recurrent component, and regression layer. In Chapter 6, Aidarbek explains the experimental setup and evaluation strategy, and presents the obtained results together with their interpretation and discussion. Chapter 7 addresses the limitations of the proposed methodology, while Chapter 8 concludes the thesis by summarizing the main findings and outlining possible directions for future research and improvement of the method.

While working on this Senior Thesis, Aidarbek Papakov demonstrated strong theoretical insight and a deep understanding of mathematical foundations and their applications in modern machine learning. He combined ideas from signal processing, deep learning, and seismology to design, formalize, and evaluate an innovative approach to earthquake magnitude estimation in earthquake early warning systems. The work is original, technically rigorous, and highly relevant for current and future developments in seismic monitoring and early warning technologies.

In my opinion, the Senior Thesis “Earthquake Magnitude Estimation from Early P-Wave Observations Using a Machine Learning Approach” fully meets the academic requirements for undergraduate research. Upon a successful defense, Aidarbek Papakov deserves an excellent grade and the awarding of the Degree of Bachelor of Arts in Applied Mathematics and Informatics.

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